

Introduction

In this assignment, I will be developing and testing a mobile phone app suitable for children below 3.

Asset Log (finished)

Description of Asset	File Name	Source	Modifications	Where used
App logo showing "Learn the Alphabet"	logo.png	Original	Adjusted the font size to make it bold for the child to see the logo easier	Start screen
Splash image of the alphabet title	splash1.png	Google Drawings		Start Screen
Letter A	a.png	Google Drawings		Letter A
Letter B	b.png	Google Drawings		Letter B
Letter C	c.png	Google Drawings		Letter C
Letter D	d.png	Google Drawings		Letter D
Letter E	e.png	Google Drawings		Letter E
Letter F	f.png	Google Drawings		Letter F
Completed	splash2.png	Google		Finish Screen

splash screen		Drawings		
Image of apple	apple.png	Google Emojis		Letter A
Image of bus	bus.png	Google Emojis		Letter B
Image of cat	cat.png	Google Emojis		Letter C
Image of dog	dog.png	Google Emojis		Letter D
Image of elephant	elephant.png	Google Emojis		Letter E
Image of fart cloud	fart.png	Google Emojis	Removed the fart screen and assets because it may be deemed "unsuitable for children"	Letter F
Start button	start.png	Google Drawings		Start Screen
Next Button	next.png	Google Drawings		Letter A-F
Back Button	back.png	Google Drawings		Letter A-F & Finish Screen
Restart Button	restart.png	Google Drawings		Finish Screen
Sound of apple biting	apple.mp3	Freesound		Letter A
Sound of bus honking	bus.mp3	Freesound		Letter B
Sound of cat meowing	cat.mp3	Freesound		Letter C
Sound of dog barking	dog.mp3	Freesound		Letter D
Sound of elephant	ele.mp3	Freesound		Letter E

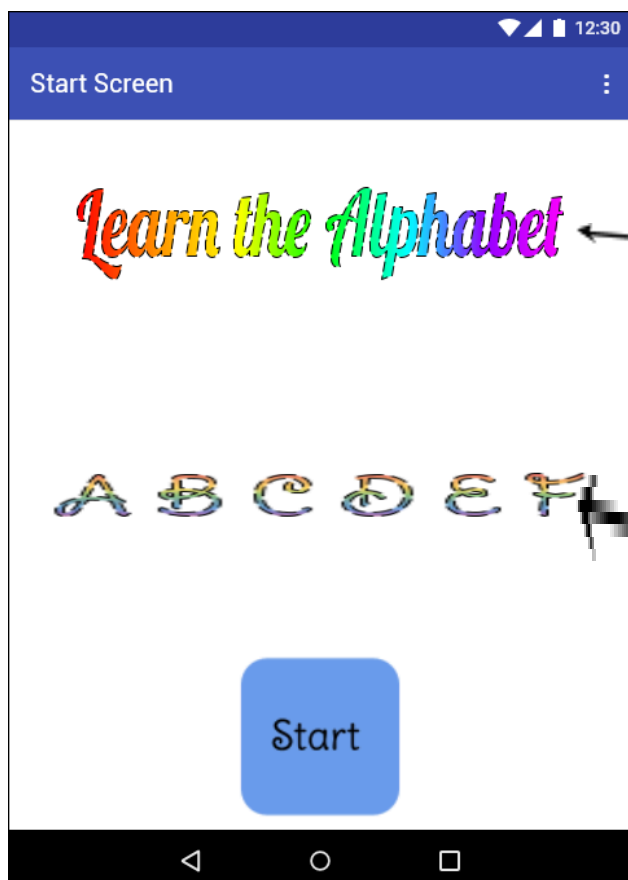
trumpeting				
Sound of fart	fart.mp3	Freesound	Removed the fart screen and replaced with fish because it may be deemed “unsuitable for children”	Letter F
Sound of fish	fish.mp3	Freesound		Letter F
TTS Start Sound	sound1.flac	NaturalReader		Start Screen
TTS Finish Sound	sound2.flac	NaturalReader		Finish Screen
People Cheering	yay.wav	Freesound		Finish Screen
TTS Letter A Sound	a.mp3	TTSFree		Letter A
TTS Letter B Sound	b.mp3	TTSFree		Letter B
TTS Letter C Sound	c.mp3	TTSFree		Letter C
TTS Letter D Sound	d.mp3	TTSFree		Letter D
TTS Letter E Sound	e.mp3	TTSFree		Letter E
TTS Letter F Sound 1	f_a.mp3	TTSFree		Letter F
TTS Letter F Sound 2	f_b.mp3	TTSFree		Letter F
Clear Button	clear.png	Google Drawings		Letter A-F

Updated Letter A	aup.png	Google Drawings		Letter A
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Completed Screens

Screen 1: Start Screen (renamed to Screen1)

This is my introduction screen. This is where the child will be introduced to the app where they will learn the alphabets A-F with a friendly-like touch to it. It includes all of the necessities for the suitable audience which are primarily children that are 3 and below.



This logo design suits the targeted audience because children tend to love colourful and flashy images which enlightens their mood and make them willingly use the app. It also meets the user requirement and purpose of the app because its demonstrating to us that this is an app to learn the alphabet.

This splash image helps the child understand how many alphabets this app will contain. Although this app doesnt contain all of the alphabet, it will still benefit the intended use of the app and make it suitable for the targeted audience. When you click on the splash image, a text-to-speech sound plays "Lets learn the Alphabet together" which makes the child happy to learn the alphabet and makes accessibility easier for those who are blind.

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when Button1 .Click
do open another screen screenName LetterA

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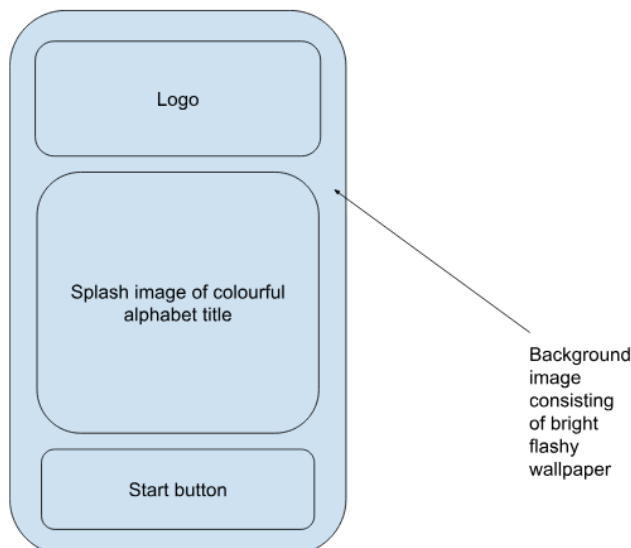
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when Image2 .Click
do call Sound1 .Play

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The first block above, makes the “Start button” function to make it go into the first alphabet, which is the letter A. The simplicity of this function makes usability better and enhances the maintainability of the screen.

The second function is where the text-to-speech sound plays when you click on the image. The reason why I have chosen this function is because the function helps the child get entertained and make it accessible for the children who are visually impaired.

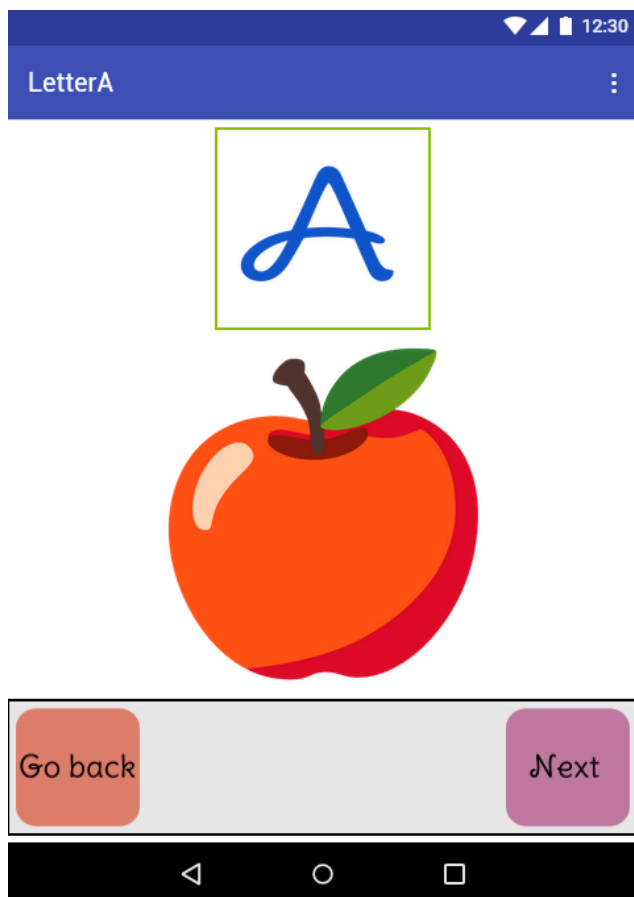


This is the original design of the screen. Although I could make the button wider to make accessible easier for children, the issue is finding the right width and size of the button to make it fit properly into the screen. Including the background wallpaper behind this screen, would be so useful for children and they would get

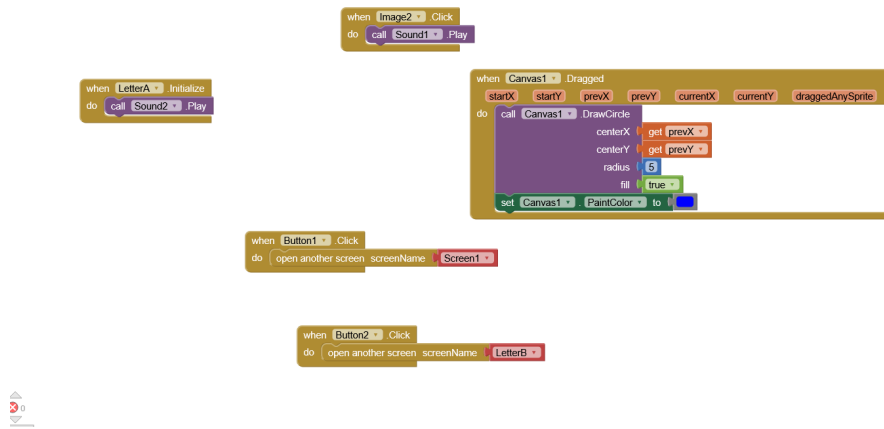
entertained. The issue is, I cannot find a good wallpaper that children will get excited about. This in turn makes the app boring and usability a little bit bad. However, the logo, start button and splash image are all in the actual app. The alternative solutions said that I could add a quit button to make it easier for them to leave the app, but since the App Inventor screen already has a quit button, there's no need.

Screen 2-7: Letter A-F

The next 5 screens show us the alphabet ranging from letter A to F. All of the final designs on these 5 screens meet the consistency of each screen, fits the user's requirements and purposes and includes code which is maintainable, easy to read and make it accessible for anyone developing this app. For this assignment, I will be cutting down to one example of the design, code and plan because since the original design is copy and pasted, it would be tedious to do each one every time.



In this example, I will be using the Letter A. It includes a very simplistic design into the screen, hence the usability and accessibility is better for children. This screen shows a very formable and simple design of the letter A, which can be drawn as a canvas with the color being blue. (the colors differ for each letter, so letter B is red, C is pink, D is yellow, E is orange and F is green)

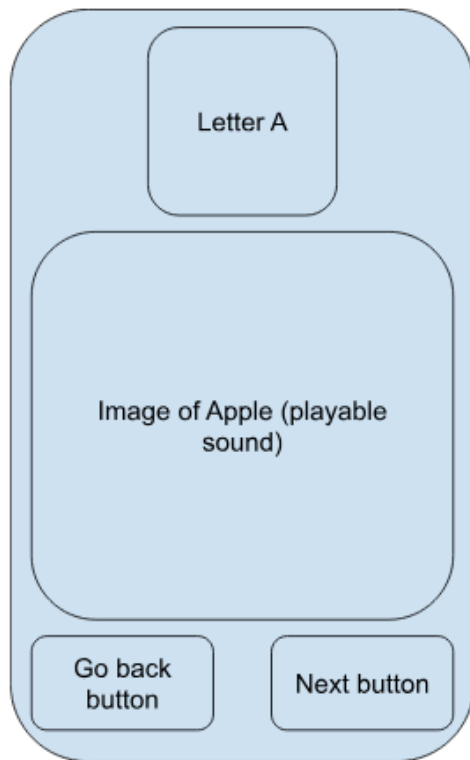


The first block at the very top, plays the sound that demonstrates of someone biting into an apple. Again, it differs from the other letters. Anyway, this helps with usability because it teaches the child to know what this object sounds like in the real world. That is also linked with the 4 other letters sounding all different. Coding wise, maintainability is a breeze for me and the developers because it is an very easy block for anyone to read.

Moving to the block on your left, this block makes the TTS speak about the item that is included on the screen and says the letter included on the screen. This in turn helps with accessibility, fits the user requirements and purpose of the app, and makes the intended audience learn what the sound says.

The big line of code to your right is the function for your canvas. This block shows when you drag your finger across the letter, the paint begins following the line and making the child learn how to write letters too. This fits the user requirements and purposes of the app because writing is a big part of your life and for a child, they need to learn how to write properly, so this canvas helps make the child learn how to write letters. The big line of code too is also apart of the predefined code i have used on Assignment 2.

And finally these 2 at the bottom, the first one on the bottom top, it makes the screen go back to the start, or any letter before the screen, and the very bottom, takes them to another screen. These 2 help with usability and fits the user requirements because it easily allows the user to get redirected to another letter easily.



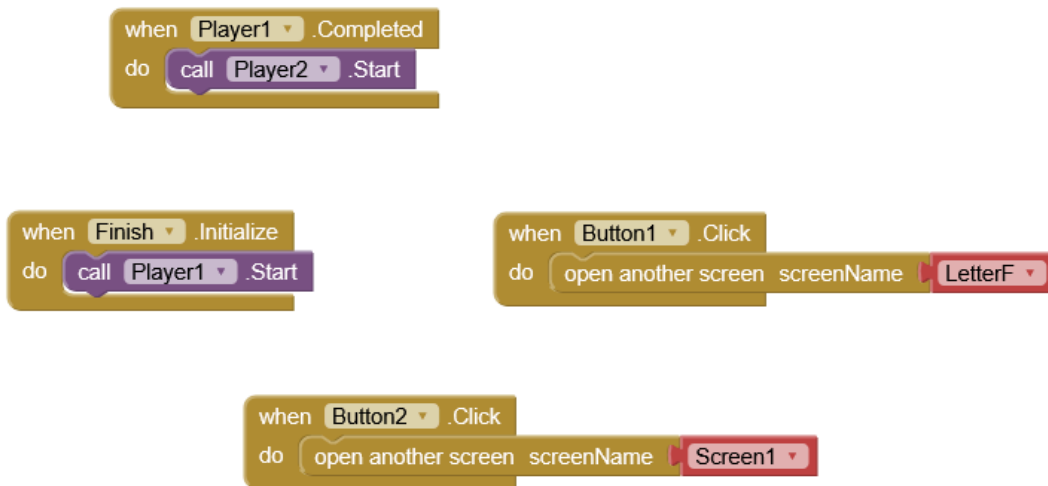
Now there is not really much to say about the plan because it has been matched perfectly with the final product, only a slight fix is that the buttons are wider than the regular square buttons. The plan has been copy and pasted for the 6 other letters too.

Screen 8: Finish Screen

The final screen. This final screen shows to us that we are finished with the app and that we have learnt the 6 alphabets.

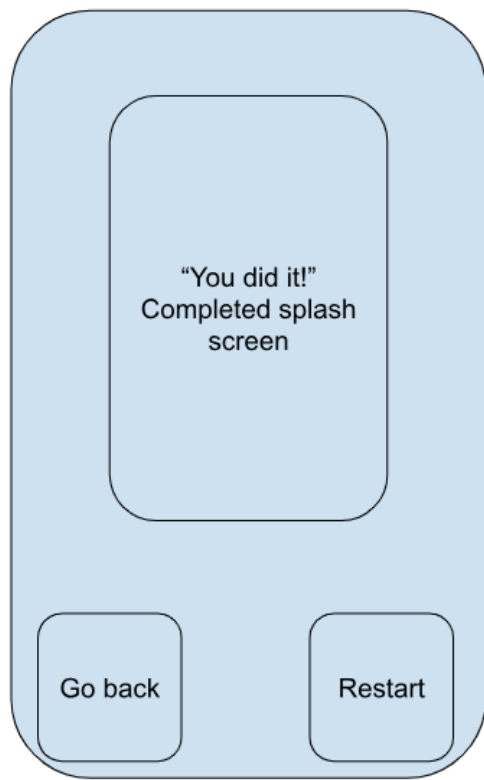


The finished screen shows to us that there is a splash image, a go back button and a restart button. With the simplicity of the finished screen, it is suitable for the intended audience as it includes a colourful text at the end, and a easy-to-read text for the splash image. Go back and restart button are displayed properly for the child to read and make the user requirements fit for the final screen. Behind the scenes, there are 2 sounds, one of them is a text-to-speech sound of someone saying “You did it” followed by people cheering. This shows to us that without looking at the screen, we have finished the alphabet and it helps the visually impaired that theyre finished with the alphabet.



The 2 call blocks on your left are your sounds. When the finish screen is initialized, the first text to speech plays. After the first text-to-speech sound is finished, the other one is played afterwards, in a non-simultaneous loop and making the screen more interesting with the new function. The reason why we have picked the Player component on the finish screen is because it includes that function where you are able to play a sound after another, when the original Sound function doesn't have. Although it doesn't really help with the maintainability of the code because it was quite tedious to use, the usability is slightly better and also it is automated, so you don't need to click on the image to play the next sound afterwards.

These 2 blocks on the right are your generic go back and restart button, which improves usability and user experience which has already been discussed.



The plan designated for the finish screen has been matched, so no comment.

Screen Recording

This is a screen recording of the Alphabets app.

Completed Test Log

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
1	Start Screen	App load and Display	Open app	Start screen displays splash image (splash1.png), image of logo (logo.png), displays the start button icon (start.png).	Outcome meets expectations.	Splash image is changed to make it more visible from the original design.
2	Start Screen	Navigation (start button)	Press start button	Letter A screen is displayed after pressed.	Outcome meets expectations.	
3	Start Screen	TTS Sound Functionality	Press the splash image to play TTS	TTS sound plays "Lets learn the Alphabet together." when pressing.	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
4	Letter A	Letter A Screen Display (UI)	System Load or Transition to screen	image of apple (apple.png), letter a (a.png), next and back button is visible. Canvas is set to color blue.	Outcome meets expectations.	
5	Letter A	Drawing Functionality	Trace the image of an apple with an finger (input)	A blue line is drawn on the canvas.	Outcome meets expectations.	
6	Letter A	Sound Functionality	Plays the sound function when pressing the image (apple.png)	An sound of someone biting to an apple plays (apple.mp3) heard clearly and loud enough.	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
7	Letter A	TTS Sound Functionality	Plays the TTS sound function when screen is initialized	An TTS sound saying “A is for Apple” heard clearly and loud enough. (a.mp3)	Outcome meets expectations.	
8	Letter A	Navigation	Press next button (next.png)	Letter B screen is displayed.	Outcome meets expectations.	
9	Letter A	Navigation	Press back button (back.png)	Start screen is displayed.	Outcome meets expectations.	
10	Letter B	Letter B Screen Display (UI)	System Load or Transition to screen	image of a bus (bus.png), letter b (b.png), next and back button is visible. Canvas is set to color red.	Outcome meets expectations.	
11	Letter B	Drawing Functionality	Trace the image of an bus with	A red line is drawn on the	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
			an finger (input)	canvas.		
12	Letter B	Sound Functionality	Plays the sound function when pressing the image (bus.png)	An sound of someone honking a horn on a bus plays (bus.mp3) heard clearly and loud enough.	Outcome meets expectations.	
13	Letter B	TTS Sound Functionality	Plays the TTS sound function when screen is initialized	An TTS sound saying "B is for Bus" heard clearly and loud enough. (b.mp3)	Outcome meets expectations.	
14	Letter B	Navigation	Press next button (next.png)	Letter C screen is displayed.	Outcome meets expectations.	
15	Letter B	Navigation	Press back button (back.png)	Start screen is displayed.	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
16	Letter C	Letter C Screen Display (UI)	System Load or Transition to screen	image of a cat (cat.png), letter c (c.png), next and back button is visible. Canvas is set to color pink.	Outcome meets expectations.	
17	Letter C	Drawing Functionality	Trace the image of an cat with an finger (input)	A pink line is drawn on the canvas.	Outcome meets expectations.	
18	Letter C	Sound Functionality	Plays the sound function when pressing the image (cat.png)	An sound of a cat meowing plays (cat.mp3) heard clearly and loud enough.	Outcome meets expectations.	
19	Letter C	TTS Sound Functionality	Plays the TTS sound function when screen is initialized	An TTS sound saying "C is for Cat" heard clearly and loud enough.	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
				(c.mp3)		
20	Letter C	Navigation	Press next button (next.png)	Letter D screen is displayed.	Outcome meets expectations.	
21	Letter C	Navigation	Press back button (back.png)	Start screen is displayed.	Outcome meets expectations.	
22	Letter D	Letter D Screen Display (UI)	System Load or Transition to screen	image of a dog (dog.png), letter d (d.png), next and back button is visible. Canvas is set to color yellow.	Outcome meets expectations.	
23	Letter D	Drawing Functionality	Trace the image of an apple with an finger (input)	A yellow line is drawn on the canvas.	Outcome meets expectations.	
24	Letter D	Sound Functionality	Plays the sound function when	An sound of some dog	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
			pressing the image (dog.png)	barking plays (dog.mp3) heard clearly and loud enough.		
25	Letter D	TTS Sound Functionality	Plays the TTS sound function when screen is initialized	An TTS sound saying “D is for dog” heard clearly and loud enough. (d.mp3)	Outcome meets expectations.	
26	Letter D	Navigation	Press next button (next.png)	Letter E screen is displayed.	Outcome meets expectations.	
27	Letter D	Navigation	Press back button (back.png)	Start screen is displayed.	Outcome meets expectations.	
28	Letter E	Letter E Screen Display (UI)	System Load or Transition to screen	image of an elephant (ele.png), letter e (e.png), next and back	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
				button is visible. Canvas is set to color orange.		
29	Letter E	Drawing Functionality	Trace the image of an elephant with an finger (input)	A orange line is drawn on the canvas.	Outcome meets expectations.	
30	Letter E	Sound Functionality	Plays the sound function when pressing the image (ele.png)	An sound of an elephant trumpeting plays (ele.mp3) heard clearly and loud enough.	Outcome meets expectations.	
31	Letter E	Navigation	Press next button (next.png)	Letter F screen is displayed.	Outcome meets expectations.	
32	Letter E	Navigation	Press back button (back.png)	Letter D screen is displayed.	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
33	Letter E	TTS Sound Functionality	Plays the TTS sound function when screen is initialized	An TTS sound saying “E is for elephant” heard clearly and loud enough. (e.mp3)	Outcome meets expectations.	
34	Letter F	Letter A Screen Display (UI)	System Load or Transition to screen	image of a fish (fish.png), letter f (f.png), next and back button is visible. Canvas is set to color green.	Outcome meets expectations.	Changed fart to fish to make it more child-friendly.
35	Letter F	Drawing Functionality	Trace the image of an apple with a finger (input)	A green line is drawn on the canvas.	Outcome meets expectations.	
36	Letter F	Sound Functionality	Plays the sound function when pressing the image (fish.png)	An sound of a fish (fish.mp3) heard clearly and loud	First outcome meets expectations. Second	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
				enough. Next sound (f_b.mp3) plays after the TTS sound plays	Outcome meets expectations.	
37	Letter F	Navigation	Press next button (next.png)	finish screen is displayed.	Outcome meets expectations.	
38	Letter F	TTS Sound Functionality	Plays the TTS sound function when screen is initialized (f_a.mp3)	An TTS sound saying "F is for fish" heard clearly and loud enough.	Outcome meets expectations.	
39	Letter F	Navigation	Press back button (back.png)	Letter E screen is displayed	Outcome meets expectations.	
40	Finish Screen	Finished screen display (UI)	System Load or Transition to Screen	Completed splash screen displays, showing "you did it!" on the splash image	Outcome meets expectations.	

Test Number	Screen	Description	Test Data	Expected Outcome	Actual Outcome	Notes
				(splash2.png), back button and restart image shows		
41	Finish Screen	Navigation (restart)	Press restart button (restart.png)	Going all the way back to the Start screen	Outcome meets expectations.	
42	Finish Screen	Navigation (go back)	Press back button (back.png)	Letter F screen is displayed.	Outcome meets expectations.	
43	Finish Screen	TTS Sound Functionality	Plays the TTS sound function when screen is initialized	An TTS sound saying "You did it!" heard clearly and loud enough.	Outcome meets expectations.	
44	Finish screen	Sound Functionality	Plays the sound after the first TTS sound was initialized	An sound of people cheering (yay.wav) heard loud and clear.	Outcome meets expectations.	